Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The cathode of a photoelectric cell is changed such that the work function changes from W_1 to W_2 $(W_2 > W_1)$. If the current before and after changes are I_1 and I_2, all other conditions remaining unchanged, then (assuming $hv > W_2$)

(A) $I_1 = I_2$

(B) $I_1 < I_2$

(C) $I_1 > I_2$

(D) $I_1 < I_2 < 2I_1$

Q.2

In a photo-emissive cell, with exciting wavelength λ, the maximum kinetic energy of electron is K. If the exciting wavelength is changed to $3\lambda/4$ the kinetic energy of the fastest emitted electron will be

(A) $3\ \mathrm{K}/4$

(B) $4\ \mathrm{K}/3$

(C) less than $4\ \mathrm{K}/3$

(D) greater than $4\ \mathrm{K}/3$

Q.3

A photoelectric cell is illuminated by a point source of light $1\ \mathrm{m}$ away. When the source is shifted to $2\ \mathrm{m}$ then

(A) number of electrons emitted is a quarter of the initial number

(B) each emitted electron carries one quarter of the initial energy

(C) number of electrons emitted is half the initial number

(D) each emitted electron carries half the initial energy

Q.4

The potential difference applied to an X-ray tube is $5\mathrm{kV}$ and the current through it is $3.2\ \mathrm{mA}$. Then the number of electrons striking the target per second is

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(A) 2×10^{16}

(B) 5×10^{6}

(C) 1×10^{17}

(D) 4×10^{15}

Q.5

A horizontal cesium plate (work function $= 1.9\ \mathrm{eV}$) is moved vertically downward at a constant speed V in a room full of radiation of wavelength $250\ \mathrm{nm}$ and above. The minimum value of v so that the vertically upward component of velocity is nonpositive for each photoelectron.

(A) $1.04 \times 10^{6}\ \mathrm{ms}^{-1}$

(B) $2.03 \times 10^{7}\ \mathrm{ms}^{-1}$

(C) $4.03 \times 10^{8}\ \mathrm{ms}^{-1}$

(D) $5.11 \times 10^{9}\ \mathrm{ms}^{-1}$

Q.6

A particle having mass $1.67 \times 10^{-27}\ \mathrm{kg}$ moving at a speed v absorbs a photon of wavelength 122 nm and stops. The value of v is

(A) $3.25\ \mathrm{ms}^{-1}$

(B) $1.28\ \mathrm{ms}^{-1}$

(C) $0.82\ \mathrm{ms}^{-1}$

(D) $0.33\ \mathrm{ms}^{-1}$

Q.7

Electrons with energy $80\mathrm{keV}$ are incident on the tungsten target of an X-ray tube. K -shell electrons of tungsten have $72.5\mathrm{keV}$ energy. X rays emitted by the tube contain only

(A) a continuous X-ray spectrum (Bremsstrahlung) with a minimum wavelength of $0.155\tilde{A}\ldots$

(B) a continuous X-ray spectrum (Bremsstrahlung) with all wavelengths

(C) the characteristic X-ray spectrum of tungsten

(D) a continuous X-ray spectrum (Bremsstrahlung) with a minimum wavelength of $0.155\tilde{A}\ldots$ and the characteristic X-ray spectrum of tungsten.

Q.8

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An X-ray tube is operated at $15\mathrm{kV}$. Calculate the upper limit of the speed of the electrons striking the target.

(A) $7.26 \times 10^7\ \mathrm{m/s}$

(B) $7.62 \times 10^7\ \mathrm{m/s}$

(C) $7.62 \times 10^7\ \mathrm{cm/s}$

(D) $7.26 \times 10^9\ \mathrm{m/s}$

Q.9

The threshold wavelength of the tungsten is 2300 $\tilde{A}\ldots$. If ultraviolet light of wavelength $1800\tilde{A}\ldots$ is incident on it, then the maximum kinetic energy of photoelectrons would be about

(A) $1.49\mathrm{eV}$

(B) $2.2\mathrm{eV}$

(C) $3.0\mathrm{eV}$

(D) $5.0\mathrm{eV}$

Q.10

Two identical photocathodes receive light of frequencies f_1 and f_2. If the velocites of the photo electrons (of mass m) coming out are respectively
v_1 and v_2, then

(A) $v_1^2 - v_2^2 = \frac{2h}{m}\left(f_1 - f_2\right)$

(B) $v_1 + v_2 = \left[\frac{2h}{m}\left(f_1 + f_2\right)\right]^{1/2}$

(C) $v_1^2 + v_2^2 = \frac{2h}{m}\left(f_1 + f_2\right)$

(D) $v_1 - v_2 = \left[\frac{2h}{m}\left(f_1 - f_2\right)\right]^{1/2}$

Q.11

When photons of wavelength λ_1 are incident on an isolated sphere, the corresponding stopping potential is found to be V. When photons of wavelength λ_2 are used, the corresponding stopping potential was thrice that of the above value. If light of wavelength λ_3 is used then find the stopping potential for this case:

(A) $\frac{\mathrm{hc}}{\mathrm{e}}\left[\frac{1}{\lambda_3} + \frac{1}{\lambda_2} - \frac{1}{\lambda_1}\right]$

(B) $\frac{\mathrm{hc}}{\mathrm{e}}\left[\frac{1}{\lambda_3} + \frac{1}{2\lambda_2} - \frac{1}{\lambda_1}\right]$

(C) $\frac{\mathrm{hc}}{\mathrm{e}}\left[\frac{1}{\lambda_3} - \frac{1}{\lambda_2} - \frac{1}{\lambda_1}\right]$

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(D) $\frac{hc}{e}\left[\frac{1}{\lambda_3}+\frac{1}{2\lambda_2}-\frac{3}{2\lambda_1}\right]$

Q.12

Two identical photocathodes receive light of frequencies f_1 and f_2. If the velocites of the photo electrons (of mass m) coming out are respectively v_1 and v_2, then

(A) $v_1^2-v_2^2=\frac{2h}{m}(f_1-f_2)$

(B) $v_1+v_2=\left[\frac{2h}{m}(f_1+f_2)\right]^{1/2}$

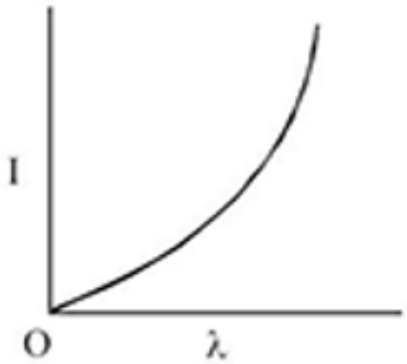
(C) $v_1^2+v_2^2=\frac{2h}{m}(f_1+f_2)$

(D) $v_1-v_2=\left[\frac{2h}{m}(f_1-f_2)\right]^{1/2}$

Q.13

The anode voltage of a photocell is kept fixed. The wavelength λ of the light falling on the cathode is gradually changed. The plate current I of the photocell varies as follows

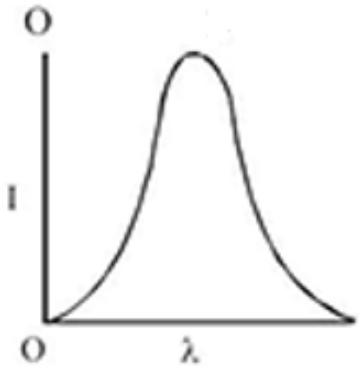
(A)



(B)

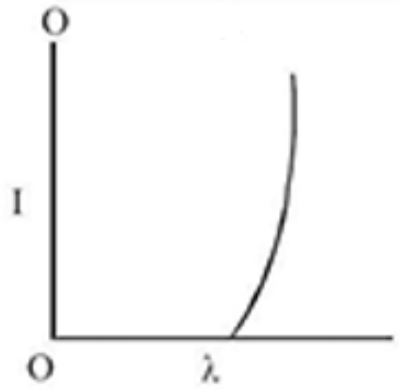


(C)



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(D)



Q.14

In a photoelectric experiment, with light of wavelength λ, the fastest electron has speed v. If the exciting wavelength is changed to $5\lambda/4$, the speed of the fastest emitted electron will become

(A) $\mathrm{v}\sqrt{\frac{5}{4}}$

(B) $v\sqrt{\frac{5}{3}}$

(C) less than $v\sqrt{\frac{5}{3}}$

(D) greater than $v\sqrt{\frac{5}{3}}$

Q.15

The cathode of a photoelectric cell is changed such that the work function changes from W_1 to W_2 ($\mathrm{W}_2 > \mathrm{W}_1$) . If the current before and after changes are I_1 and I_2, all other conditions remaining unchanged, then (assuming $\mathrm{hv} > \mathrm{W}_2$)

(A) $\mathrm{I}_1 = \mathrm{I}_2$

(B) $\mathrm{I}_1 < \mathrm{I}_2$

(C) $\mathrm{I}_1 > \mathrm{I}_2$

(D) $\mathrm{I}_1 < \mathrm{I}_2 < 2\mathrm{I}_1$

Q.16

The maximum velocity of an electron emitted by light of wavelength λ incident on the surface of a metal of workfunction ϕ is

(A) $\sqrt{\frac{2(\mathrm{hc}+\lambda\phi)}{\mathrm{m}\lambda}}$

(B) $\frac{2(\mathrm{hc}+\lambda\phi)}{\mathrm{m}\lambda}$

(C) $\sqrt{\frac{2(\mathrm{hc}-\lambda\phi)}{\mathrm{m}\lambda}}$

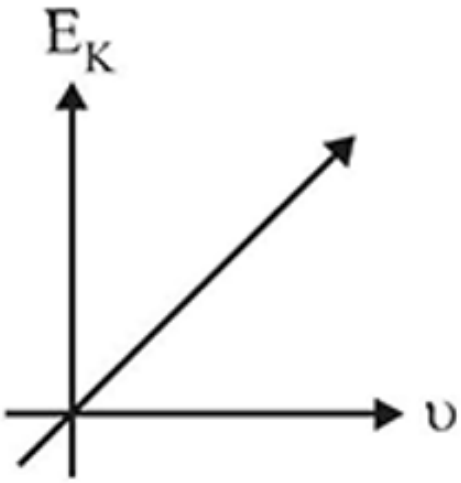
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(D) $\sqrt{\frac{2(h\lambda-\phi)}{m}}$

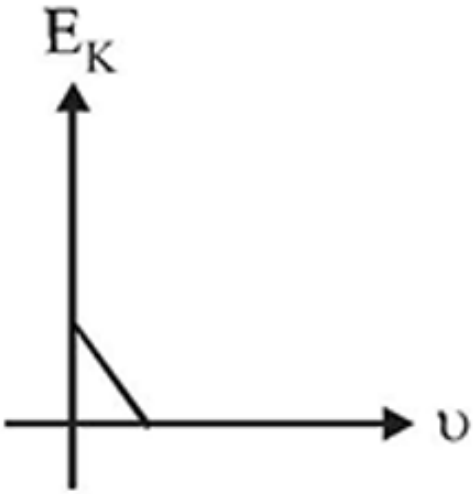
Q.17

Which one of the following graphs represents the variation of maximum kinetic energy (E_K) of the emitted electrons with frequency υ in photoelectric effect correctly?

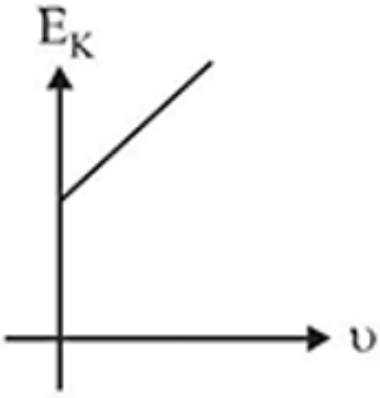
(A)



(B)

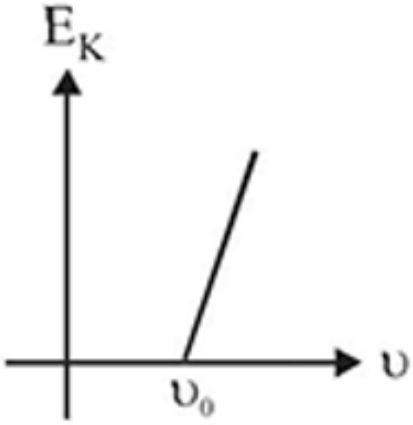


(C)



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(D)



Q.18

If E_1, E_2, E_3 are the respective kinetic energies of an electron, an alpha-particle and a proton, each having the same de-Broglie wavelength, then

(A) $\mathrm{E_1 > E_3 > E_2}$

(B) $\mathrm{E_2 > E_3 > E_1}$

(C) $\mathrm{E_1 > E_2 > E_3}$

(D) $E_1 = E_2 = E_3$

Q.19

In photoelectric effect, stopping potential for a light of frequency n_1 is V_1. If light is replaced by another having a frequency n_2 then its stopping potential will be

(A) $\mathrm{V_1 - \frac{h}{e}(n_2 - n_1)}$

(B) $\mathrm{V_1 + \frac{h}{e}(n_2 + n_1)}$

(C) $\mathrm{V_1 + \frac{h}{e}(n_2 - 2n_1)}$

(D) $V_1 + \frac{h}{e}(n_2 - n_1)$

Q.20

A certain metallic surface is illuminated with monochromatic light of wavelength λ. The stopping potential for photoelectric current for this light is $3\ \mathrm{V_0}$. If the same surface is illuminated with light of wavelength 2λ, the stopping potential is $\mathrm{V_0}$. The threshold wavelength for this surface for photo-electric effect is

(A) 4λ

(B) $\frac{\lambda}{4}$

(C) $\frac{\lambda}{6}$

(D) 6λ

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Q.1 Answer:

$I_1 = I_2$

Solution:

The work function has no effect on photoelectric current so long as $hv > W_0$. The photoelectric current is proportional to the intensity of incident light. Since there is no change in the intensity of light, hence $I_1 = I_2$.

Q.2 Answer:

greater than $4\ \mathrm{K}/3$

Solution:

No solution available

Q.3 Answer:

number of electrons emitted is a quarter of the initial number

Solution:

Power $\propto$ No. of electrons emitted (N) $P \propto \frac{1}{r^2} \Rightarrow N \propto \frac{1}{r^2}$

Q.4 Answer:

2×10^{16}

Solution:

$I = \frac{q}{t} = \frac{ne}{t}$

No. of electrons striking the target per second

$= \frac{I}{e} = 2 \times 10^{16}$

Q.5 Answer:

$1.04 \times 10^6\ \mathrm{ms}^{-1}$

Solution:

When $\lambda = 250\ \mathrm{nm}$

Energy of photon $= \frac{hc}{\lambda} = \frac{1240}{250} = 4.96\mathrm{eV}$

$\therefore\ \mathrm{K.E.} = \frac{\mathrm{hc}}{\lambda} - \mathrm{w} = 4.96 - 1.9\mathrm{eV} = 3.06\mathrm{eV}$

Velocity to be non positive for each photo electron

The minimum value of velocity of plate should be = velocity of photo electron $\therefore$ Velocity of

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photo electron $= \sqrt{2\mathrm{KE}/\mathrm{m}}$

$$= \sqrt{\frac{3.06}{9.1 \times 10^{-31}}}$$

$$= \sqrt{\frac{3.06 \times 1.6 \times 10^{-19}}{9.1 \times 10^{-31}}}$$

$= 1.04 \times 10^{6}\ \mathrm{m/sec}$

Q.6 Answer:

$3.25\ \mathrm{ms}^{-1}$

Solution:

No solution available

Q.7 Answer:

a continuous X-ray spectrum (Bremsstrahlung) with a minimum wavelength of $0.155\tilde{A}\ldots$ and the characteristic X-ray spectrum of tungsten.

Solution:

$\lambda_{\min} = \frac{hc}{E} \quad \therefore \lambda_{\min} = \frac{12400}{80\times10^{3}}\tilde{A}\ldots = 0.155\tilde{A}\ldots$

Energy of incident electrons is greater than the ionization energy of electrons in K **-shell, the K-shell electrons will be knocked off. Hence, characteristic X-ray spectrum will be obtained.**

Q.8 Answer:

$7.26 \times 10^{7}\ \mathrm{m/s}$

Solution:

The maximum kinetic energy of an electron accelerated through a potential difference of V **volt is** $\frac{1}{2}mv^{2} = \mathrm{eV}$ **∴ maximum velocity** $\mathrm{v} = \sqrt{\frac{2\mathrm{eV}}{\mathrm{m}}}$

$\mathrm{v} = \sqrt{\frac{2\times1.6\times10^{-19}\times15000}{9.1\times10^{-31}}}$

$\mathrm{v} = 7.26 \times 10^{7}\ \mathrm{m/s}$

Q.9 Answer:

$1.49\mathrm{eV}$

Solution:

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$K_{max} = hv - hv_0 = hc\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right)$
Use $hc = 1.24 \times 10^{-6}(eV)m$
$hc = 1.24 \times 10^{-6}\left(\frac{10^8}{18} - \frac{10^8}{23}\right)$
$= \frac{1.24 \times 100 \times (23-18)}{18 \times 23} = 1.49eV$

Q.10 Answer:

$v_1^2 - v_2^2 = \frac{2h}{m}(f_1 - f_2)$

Solution:

For one photocathode $hf_1 - W = \frac{1}{2}mv_1^2$
For another photo cathode $hf_2 - W = \frac{1}{2}mv_2^2$
Subtracting (ii) from (i) we get $(hf_1 - W) - (hf_2 - W) = \frac{1}{2}mv_1^2 - \frac{1}{2}mv_2^2$
$\therefore h(f_1 - f_2) = \frac{m}{2}\left(v_1^2 - v_2^2\right)$
$\therefore v_1^2 - v_2^2 = \frac{2h}{m}(f_1 - f_2)$

Q.11 Answer:

$\frac{hc}{e}\left[\frac{1}{\lambda_3} + \frac{1}{2\lambda_2} - \frac{3}{2\lambda_1}\right]$

Solution:

Stopping potential is $V_0 = \frac{hc}{\lambda e}$

$$\rightarrow V = \frac{hc}{\lambda_1 e}$$

$$\rightarrow 3\ V = \frac{hc}{\lambda_2 e}$$

$$\Rightarrow \lambda_1 = 3\lambda_2$$

Now, Stopping potential for photon of wavelength λ_3 **is** $\frac{hc}{\lambda_3 e}$ **which is also equal to**
$\frac{hc}{e}\left\{\frac{1}{\lambda_3} + \frac{1}{2\lambda_2} - \frac{3}{2\lambda_1}\right\}$

Q.12 Answer:

$v_1^2 - v_2^2 = \frac{2h}{m}(f_1 - f_2)$

Solution:

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For one photocathode

$hf_1 - W = \frac{1}{2}mv_1^2$

For another photo cathode

$hf_2 - W = \frac{1}{2}mv_2^2$

Subtracting (ii) from (i) we get

$(hf_1 - W) - (hf_2 - W) = \frac{1}{2}mv_1^2 - \frac{1}{2}mv_2^2$

$\therefore h(f_1 - f_2) = \frac{m}{2}\left(v_1^2 - v_2^2\right)$

$\therefore v_1^2 - v_2^2 = \frac{2h}{m}(f_1 - f_2)$

Q.13 Answer:



Solution:

As λ decreases, y increases and hence the speed of photoelectron increases. The chances of photo electron to meet the anode increases and hence photo electric current increases.

Q.14 Answer:

less than $v\sqrt{\frac{5}{3}}$

Solution:

Since, stopping potential is independent of distance hence new stopping potential will remain unchanged

i.e., new stopping potential $= V_0$.

Q.15 Answer:

$\mathrm{I}_1 = \mathrm{I}_2$

Solution:

The work function has no effect on photoelectric current so long as $hv > W_0$. The photoelectric current is proportional to the intensity of incident light. Since there is no change in the intensity of light, hence $I_1 = I_2$.

Q.16 Answer:

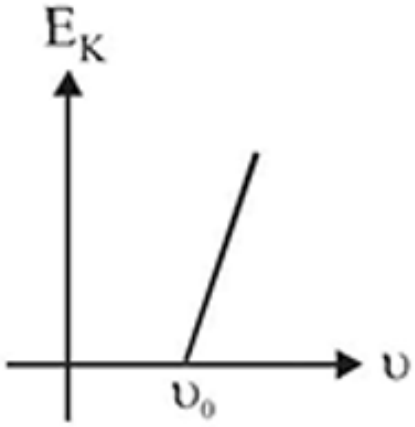
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$\sqrt{\frac{2(hc-\lambda\phi)}{m\lambda}}$

Solution:

$\frac{1}{2}mv^2 = \frac{hc}{\lambda} - \phi \Rightarrow v = \sqrt{\frac{2(hc-\lambda\phi)}{\lambda m}}$

Q.17 Answer:



Solution:

$hv - hv_0 = E_K$, **according to photoelectric equation, when** $v = v_0, E_K = 0$

Graph (d) represents $E_K - \cup$ **relationship.**

Q.18 Answer:

$E_1 > E_3 > E_2$

Solution:

According to relation, $E = \frac{1}{2}mv^2$

$\sqrt{\frac{2E}{m}} = v$

$\lambda = \frac{h}{\sqrt{2mE}}$

Because $m_1 < m_3 < m_2$

So for same $\lambda, E_1 > E_3 > E_2$

Q.19 Answer:

$V_1 + \frac{h}{e}(n_2 - n_1)$

Solution:

$$W_0 = hv_1 - eV_1$$
$$= hv_2 - eV_2$$
$$eV_2 = h(v_2 - v_1) + eV_1$$
$$V_2 = \frac{h(n_2 - n_1)}{e} + V_1$$

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Q.20 Answer:

4λ

Solution:

As we know, $eV_s = \frac{hc}{\lambda} - \Psi$

$3eV_o = \frac{hc}{\lambda} - \Psi$

$eV_o = \frac{hc}{2\lambda} - \Psi$

$3eV_o = \frac{3hc}{2\lambda} - 3\Psi$

Multiplying eqn. (2) by (3) and subtracting it from eqn.(1)

$\Psi = \frac{hc}{4\lambda}$

So, threshold wavelength,

$\lambda_{th} = \frac{hc}{\Psi} = \frac{hc}{hc/4\lambda} = 4\lambda$

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